STATISTICAL ANALYSIS & DATA INTERPRETATION

Assessment Solutions — Questions 16–20

Continuous Distributions • Sampling • Covariance & Correlation • Confidence Intervals • Outlier Analysis

Name : M. SREEVARDHAN

REG.NO : 192472405

COURSE : Fundamentals of Data Science

CODE : DSA0405

CO3-AT2: Data Interpretation

Assessment objective

Apply statistical reasoning to probability, sampling, association, estimation, and outlier detection. All calculations are shown clearly, followed by interpretation and a decision-oriented conclusion.

Executive Overview

Question	Core Concept	Key Result
16	Normal distribution / empirical rule	$P(85 < X < 130) = 81.86\%$; 145 is $+3\sigma$
17	Sampling bias vs sampling error	Online-only sampling creates coverage/selection bias; larger n does not remove systematic bias
18	Covariance vs correlation	Covariance magnitude depends on measurement units; standardized correlation is required
19	95% confidence interval	CI = [242.16, 257.84]
20	Integrated outlier case	500 is an extreme outlier; IQR and z-score both flag it; investigate before reporting

16. Continuous Distribution — Area Interpretation

Given: $X \sim N(\mu = 100, \sigma = 15)$.

(a) Proportion of values between 85 and 130

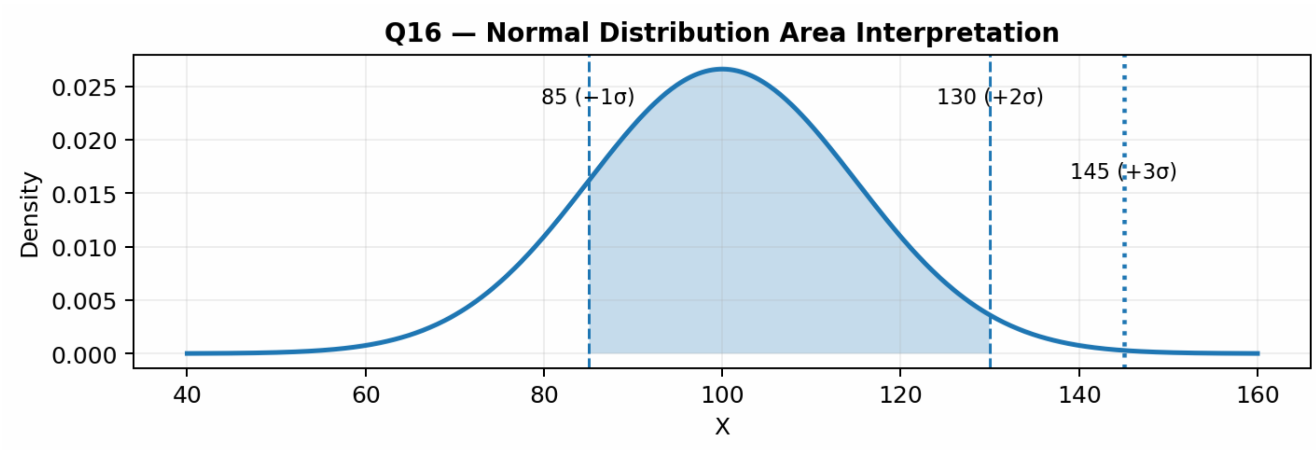
Standardize both limits: $z_1 = (85 - 100)/15 = -1.00$ and $z_2 = (130 - 100)/15 = +2.00$.

Therefore, $P(85 \le X \le 130) = P(-1 \le Z \le 2) = \Phi(2) - \Phi(-1)$.

Using the standard normal table: $\Phi(2) \approx 0.97725$ and $\Phi(-1) \approx 0.15866$. Hence, probability $\approx 0.81859 =$ 81.86%.

Answer (a)

Approximately 81.86% of observations are expected to lie between 85 and 130.



(b) Position of 145 and outlier decision

$z = (145 - 100)/15 = 3.00$. Thus, 145 is exactly 3 standard deviations above the mean.

Under the empirical rule, approximately 99.7% of normally distributed observations lie within $\pm 3\sigma$, leaving about 0.3% outside that range. A value at $+3\sigma$ is therefore unusual and should be flagged as a potential outlier.

Answer (b)

145 is $+3\sigma$ from the mean. It is a borderline/extreme value under the empirical rule and should be flagged for investigation, especially if the dataset contains additional observations beyond it.

17. Sampling Bias vs Sampling Error

Scenario: A survey estimates average household expenditure using only online panel respondents. The sample mean is ₹18,000, while the known population mean is ₹14,500.

(a) Is the discrepancy sampling error or sampling bias?

The discrepancy is better explained by sampling bias, specifically coverage/selection bias. The sampling frame excludes households that are not represented in the online panel, so the sampled group may systematically differ from the target population in income, technology access, age, location, or spending behavior.

Answer (a)

Sampling bias is the primary concern. A large difference between sample and population means is not, by itself, proof of bias, but the online-only design provides a clear systematic mechanism for producing a biased estimate.

(b) Would increasing the sample size fix the problem?

No. Increasing the sample size mainly reduces random sampling error. If the sampling frame remains restricted to online panel respondents, a very large sample can estimate the mean of that biased frame more precisely while still missing systematically excluded households.

Answer (b)

A larger sample does not cure systematic sampling bias. The remedy is to improve the sampling design—e.g., use a probability-based frame that covers the target population and apply appropriate weighting/nonresponse adjustments.

18. Covariance Sign vs Correlation Strength

Given: Pair 1 has Cov = 850 and Pair 2 has Cov = 12. Both covariances are positive.

A positive covariance tells us the variables tend to move in the same direction: when one variable increases, the other tends to increase as well. However, the numerical size of covariance is not directly comparable across different variable pairs.

The reason is that covariance depends on the units and scales of both variables. For example, measuring one variable in rupees rather than thousands of rupees changes the covariance numerically without changing the underlying relationship.

What is missing?

We need the standard deviations (or, equivalently, the standardized correlation coefficient r) for each pair. Correlation removes the effect of measurement scale and ranges from −1 to +1, making relationship strength comparable.

Measure	What it tells us	Can it compare strength across pairs?
Covariance	Direction and scale-dependent joint variation	No
Correlation (r)	Direction and standardized linear strength	Yes
Standard deviations	Required to standardize covariance	Needed to compute r

19. Estimation — Confidence Interval Reasoning

Given: n = 100, sample mean = 250, sample SD = 40. For a 95% interval, using the large-sample normal critical value 1.96:

Standard error = 40 / √100 = 4. Margin of error = 1.96 × 4 = 7.84.

Therefore, the 95% confidence interval is 250 ± 7.84 = [242.16, 257.84].

95% Confidence Interval

The estimated population mean is between 242.16 and 257.84, based on this sample and the stated 95% confidence procedure.

Interpretation: If the same population were repeatedly sampled using the same procedure and a 95% interval were constructed each time, about 95% of those intervals would contain the true population mean. The interval expresses uncertainty in the estimation procedure.

What it does NOT tell us: It does not mean there is a 95% probability that the already-fixed true population mean lies inside this particular interval. The true mean is a fixed parameter; the interval is the result of a random sampling procedure.

It also does not imply that 95% of individual observations fall between the two endpoints. A confidence interval concerns the population mean, not the spread of individual data values.

20. Integrated Case Study — Retail Daily Sales

Daily sales (₹'000) for 15 days:

120, 125, 118, 130, 122, 128, 500, 119, 124, 121, 126, 123, 117, 129, 125

20(a). Identify the outlier using IQR and z-score

Sort the data: 117, 118, 119, 120, 121, 122, 123, 124, 125, 125, 126, 128, 129, 130, 500.

Using the percentile/median-of-halves convention: Q1 = 120.5, Q3 = 127.0. Thus IQR = Q3 − Q1 = 6.5.

Lower fence = Q1 − 1.5(IQR) = 120.5 − 9.75 = 110.75. Upper fence = Q3 + 1.5(IQR) = 127.0 + 9.75 = 136.75.

The value 500 is far above the upper fence of 136.75, while all ordinary daily sales lie within the fences. Therefore, 500 is an IQR outlier.

Using the sample standard deviation, s ≈ 97.33, the z-score of 500 is z = (500 − 148.47)/97.33 ≈ 3.61.

Because $|z| > 3$, the observation is an extreme value under the common z-score rule. Both independent methods therefore identify 500 as an outlier.

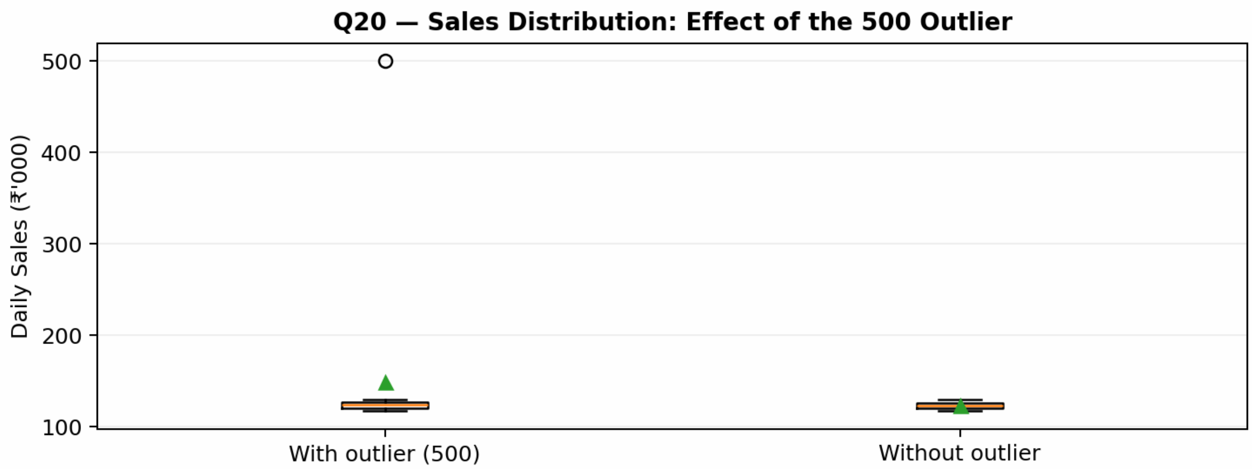
Method	Criterion	Result for 500	Decision
IQR	Outside [110.75, 136.75]	500 > 136.75	Outlier
Z-score	$|z| > 3$	$|3.61| > 3$	Extreme outlier

20(b). Mean and median with and without the outlier

With 500: mean = 148.47 (₹'000), median = 124.00 (₹'000).

Without 500: mean = 123.36 (₹'000), median = 123.50 (₹'000).

Statistic	With 500	Without 500	Change	Interpretation
Mean	148.47	123.36	25.11 (20.4%)	Highly sensitive to the extreme value
Median	124.00	123.50	0.50 (0.4%)	Much more robust to the extreme value



The mean rises dramatically because 500 contributes a very large value to the total. The median changes only slightly because it depends on the middle position after sorting and is therefore resistant to a single extreme observation.

20(c). Skewness with the outlier included

The moment coefficient of skewness for the 15 observations is approximately 3.46.

> **Distribution shape**
>
> The skewness is strongly positive. The distribution is therefore positively (right) skewed, with the long right tail created by the extreme 500 observation.

This agrees with the large separation between the mean (148.47) and median (124.00). In a strongly right-skewed distribution, unusually large values pull the mean upward more than the median.

20(d). Recommended treatment before reporting

Recommendation: INVESTIGATE the 500 value first rather than automatically removing or capping it.

- Statistical evidence is strong: 500 exceeds the IQR upper fence and has a z-score of about +3.61.
- The value materially changes the mean from 123.36 to 148.47 (₹'000), while the median changes only from 123.50 to 124.00 (₹'000).
- The included outlier produces strong positive skewness (≈ 3.46), indicating that the raw distribution is not representative of the central pattern of the other 14 days.
- However, an outlier is not automatically an error. It could represent a genuine high-sales event such as a promotion, bulk order, holiday, or major customer transaction.
- If 500 is a verified genuine event, retain it and report it transparently—preferably with robust summaries such as the median and with a separate discussion of the exceptional day.
- If 500 is confirmed to be a data-entry, measurement, or recording error, correct or remove it according to the company's data-quality policy.
- Capping should be considered only when there is a documented business/statistical policy for winsorization; it should not be used simply to make the distribution look normal.

Final decision

Overall Assessment Conclusion

Skill Demonstrated	Evidence from Q16–Q20
Probability reasoning	Converted raw values to z-scores and obtained an interval probability of 81.86%.
Sampling design judgment	Distinguished systematic sampling bias from random sampling error and explained why larger n does not remove bias.
Association interpretation	Explained why covariance magnitude cannot be compared across differently scaled variable pairs.
Statistical estimation	Constructed and interpreted a 95% CI of [242.16, 257.84].
Outlier detection & decision-making	Used both IQR and z-score evidence, quantified mean/median sensitivity, classified skewness, and recommended investigation before treatment.

Key takeaway

Strong statistical analysis does not stop at calculation. It connects numerical evidence to interpretation, limitations, data quality, and a defensible real-world decision.